



data. Almost every software these days uses a data pipeline to ensure that data can be gathered, stored, and analysed once it reaches a certain scale.

A typical pipeline consists of three stages:

The programme delivers data to the data “firehose,” which provides a streaming interface for ingesting and processing data, typically events concerning user activities. Raw data is frequently altered or supplemented before being transferred to another firehose. The most popular technologies for this are AWS Kinesis and Kafka.

Cloud storage is used to save both the raw data and the final transformed/augmented data. AWS Kinesis has a feature called “firehose” that enables saving raw data to its cloud storage (S3) incredibly simple. For analysis, the transformed/augmented data is frequently placed into a data warehouse.

You can use AWS Redshift for that and it’s very popular among startups for that reason. However, larger firms may frequently use Oracle or other proprietary warehouse systems. For analysis, a Hadoop-like NoSQL MapReduce technology may be necessary if the data sets are large enough. Another phase not depicted in the architecture diagram is importing data into the data warehouse from the app and service’s operational databases.

Cloud storage

According to AWS, “cloud storage provides a straightforward and scalable way to store, access, and share data via the Internet.” It can be used to store and access almost anything that would be stored on a local file